In-Context Gibbs: Learning from Weak Labels with Frozen Tabular Foundation Models

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Abstract—Tabular foundation models (TFMs) predict labels in context without training, but they need labelled examples. We show that a frozen TFM can learn from weak labels such as candidate sets or label proportions. Under exchangeability, the distribution of one unknown label given all the others is a posterior predictive, which is what a TFM approximates. In-Context Gibbs therefore samples the latent labels with forward passes, brings in the weak labels through their likelihood, and predicts by averaging over the samples. Its exact variants target the Bayes posterior when the TFM is exact, and a model of the TFM's response to noisy context labels explains why the start of a chain can matter. Three pre-registered studies test the method. On 28 new datasets a single chain with TabICL v2 outperformed specialised learners for label proportions (+0.052 AUC over the registered comparator) and for partial labels (+0.012 accuracy). On 20 further datasets with four TFMs, averaging chains from several starts improved on a single chain, and averaging over TFMs improved on a TFM drawn at random and matched the best TFM of each dataset. The multi-TFM average also beat untuned tree-based self-training and partial-label nearest neighbours on every dataset of their settings.

Index Terms—Weak supervision, partial label learning, learning from label proportions, tabular foundation models, in-context learning, Gibbs sampling.

1 Introduction

Labels in tabular data often arrive weak: an annotator who cannot decide records a *candidate set*, and data released under privacy constraints give only the *proportion* of each label within groups. Each form of weak supervision has its own family of learners, trained per task with a loss designed for that observation [1–5].

Tabular foundation models (TFMs) such as TabPFN [6, 7] and TabICL [8, 9] predict the label of a query point in one forward pass, conditioning on a labelled context, without training. Weak labels do not fit this interface: the context must be labelled. A learner given one plausible label per point uses only part of what candidate sets and label proportions say (Proposition S5, computed for a logistic model).

A TFM approximates a Gibbs sampler for unknown labels.: A TFM approximates the posterior predictive distribution (PPD) of its prior [10]. Under the exchangeability built into that prior, the conditional distribution of one unknown label given all the others is precisely such a PPD, with the other points and their current labels as the context (Lemma 1). A forward pass of a frozen TFM is therefore, for an exact TFM, the full conditional of a Gibbs sampler over the latent labels, and approximately so for a real one; weak labels enter only through their likelihood. In-Context Gibbs samples the unknown labels block by block with forward passes, respecting candidate sets or bag counts, and predicts by averaging the TFM's predictions over the samples (Section 4). No weight of the TFM is updated; only a chain's start may use a trained model.

Evidence from three pre-registered studies.: Each study fixed its hypotheses, comparators and tests before its evaluation data were touched, and each was designed around what the previous one found. Study 1, with TabPFN v2, showed large gains over the naive use of the TFM and that the results depend on the TFM: under heavy ambiguity TabPFN v2's chains could stall, while TabICL v2's did not. Study 2 changed only the TFM, to TabICL v2, on 28 new datasets (Table 1): a single chain outperformed the specialised neural and linear baselines for label proportions, by +0.052 AUC over the registered comparator and +0.041 over the strongest one, and for partial labels, by +0.012 and +0.012 accuracy over tuned CC. To study when a chain stalls we use a *response map*: the accuracy of the TFM's relabelling as a function of the accuracy of the labels in its context. In a model in which this map describes the TFM, it bounds where the sampler can go on instance-level weak labels (Theorem 3): with two attractors the start matters. We therefore average chains from several starts, one of them a tuned specialised learner, and then average over TFMs. Study 3 tested both steps on 20 further datasets with four TFMs (Table 2): both hypotheses were confirmed. Several starts improved on one, and the multi-TFM average improved on a TFM drawn at random and matched the best TFM of each dataset without knowing which it is. A supplementary analysis on the same splits (Section 6.5) added non-neural baselines, the naive TFM and hard-EM for all four TFMs, and chains run for 100 sweeps: the multi-TFM average beat both untuned non-neural baselines, and longer chains kept improving accuracy (+0.053 on average). Section 6.6 turns these results into a recommendation ordered by cost.

Contributions.: (1) In-Context Gibbs, a sampler over latent labels built from a frozen TFM, for instance-level and group-level weak labels, with averages over starts and over TFMs. (2) Theory: exactness with an exact TFM, stability with an approximate one, and, for the default sampler on instance-level weak labels with unsubsampled contexts and under a homogeneity assumption, floors and ceilings of the

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label accuracy that hold with high probability. (3) Three pre-registered studies on 72 datasets, three real partial-label datasets and a registered supplementary analysis, with every registered outcome reported. Proofs, protocols and further results are in the supplementary material (Appendix A onwards).

2 Related work

Weak labels. Specialised learners are trained with a loss tailored to the observation, for partial labels [1, 2, 11] and for label proportions [3, 4, 12, 13]; older non-neural learners use nearest neighbours for partial labels [14, 15] and calibrated or proportion-constrained SVMs for label proportions [16, 17]; for label proportions the number of samples needed grows linearly with the bag size [18]. Unified frameworks treat many types with one EM scheme [5, 19, 20], assuming conditionally independent predictions. We replace the training of a task-specific predictor by inference with a frozen TFM: latent labels are sampled jointly, and their dependence is modelled through the TFM's context. **Tabular foundation models.** Prior-fitted networks approximate the PPD of a prior over datasets [10]; later models extend their range [21] or pretrain on real tables [22], and theory studies when transformers learn PPDs in context [23]. Their behaviour with imperfect or missing context labels has been studied [24, 25], and iterating TabPFN on its own soft pseudo-labels has been used for semi-supervised learning [25]; other work pretrains new in-context models for one weak-label type [26–28]. We keep a general TFM frozen and condition on the weak labels through their likelihood. **Gibbs sampling with learned conditionals.** TabPFN's predictive has been used for single-pass imputation [29, 30]. Drawing labels from a PFN's predictive to resample future data underlies martingale posteriors for PFNs [31, 32], and iterating in-context learning on its own outputs samples from a model's prior [33]. Iterative imputation with chained conditional models [34] and amortised Gibbs samplers [35] are closest in spirit. Our sampler conditions each block of latent labels on the current labels of all other points and uses the weak-label likelihood; its exact variants (single-site, swap-MH, block-MH) have guarantees for incompatible conditionals, while the default uncorrected-block sampler has only the prediction-level analysis of Proposition S3 (Table S22, Appendix H). Redrawing a block given the others' current labels resembles Hogwild and asynchronous Gibbs sampling, whose bias and mixing have been analysed [36, 37]; the variant we call stochastic EM, which predicts from a single draw of the labels, follows Celeux and Diebolt [38], and alternating draws of latent data and parameters is data augmentation [39]. That iterative procedures over latent labels converge only from some starts is known for EM [40]; our floors and ceilings (Theorem 3) are a counterpart, in a model of the TFM, for the sampler on instance-level weak labels.

3 Problem setup

We observe features $X = (x_1, \dots, x_n)$ whose labels $y \in [C]^n$ are latent. As in the prior-fitted networks behind TFMs [10], we assume an exchangeable prior: $\phi \sim p(\phi)$ and $(x_i, y_i) \overset{\text{iid}}{\sim} p(x, y \mid \phi)$, with a test pair (x_*, y_*) from the same mechanism. For a labelled context D, the *posterior predictive distribution* (PPD) of the prior is $p(y_* \mid x_*, D)$; a TFM returns an approximation $\tilde{p}(y_* \mid x_*, D)$ in one forward pass. Instead of y we observe weak labels W with a known likelihood $\ell(y) = p(W \mid X, y)$, independent of (x_*, y_*) given (X, y) (A0). For *partial labels* (PLL) each point has a candidate set $S_i \ni y_i$ and $\ell(y) = \prod_i \mathbb{1}[y_i \in S_i]$, under the standard assumption that the probability of a candidate set does not depend on which member is true. For *label proportions* (LLP) the points form disjoint bags B_g with observed label counts w_g, and $\ell(y) = \prod_g \mathbb{1}[\text{counts}(y_{B_g}) = w_g]$. PU data, label noise and semi-supervised learning are further instance-level cases, $\ell(y) = \prod_i \ell_i(y_i)$ (Appendix M). The targets are the posterior over latent labels and the predictive,

$$\begin{aligned} \pi(y) = p(y \mid X, W) &\propto p(y \mid X)\, \ell(y), \\ p(y_* \mid x_*, X, W) &= \textstyle\sum_y \pi'(y)\, p(y_* \mid x_*, (X, y)), \end{aligned} \tag{1}$$

with $\pi'(y) = p(y \mid x_*, X, W)$. The analysis assumes (A1) an exact TFM, $\tilde{p} = p$, on every queried context; (A2) a positive PPD; and (A3) $\sum_y p(y \mid X)\ell(y) > 0$. Theorem 2 removes (A1).

Algorithm 1 In-Context Gibbs, one chain (block version; LLP uses bag-blocks). No TFM weight is updated.

Require: features X, weak labels W with model ℓ, frozen TFM $\tilde{p}$; K blocks, T sweeps, burn-in T_0, S samples

1: $P_0 \leftarrow$ warm start; draw $y \sim P_0 \cdot \ell$ ▷ LLP: conditioned on bag counts
2: **for** $t = 1, \dots, T$ **do**
3: partition all points (LLP: all bags) at random into blocks $B_1, \dots, B_K$
4: **for** $k = 1, \dots, K$ **do**
5: $Q \leftarrow \tilde{p}(\cdot \mid x_i, D_{-B_k})$ for all $i \in B_k$ ▷ one forward pass
6: PLL: draw $y_i \propto Q_i\, \ell_i$ for free $i \in B_k$; LLP: draw $y_{B_g} \sim \prod_{i \in B_g} Q_i$ given counts w_g, for $B_g \subseteq B_k$
7: *(optional)* accept the new y_{B_k} with the MH probability of Proposition S2(a)
8: **return** $\hat{p}(y_* \mid x_*) = \frac{1}{S} \sum_s \tilde{p}(y_* \mid x_*, (X, y^{(s)}))$ over S evenly spaced states $y^{(s)}$ with $t > T_0$

4 In-Context Gibbs

Algorithm 1 gives one chain; the paragraphs below explain its steps.

The full conditional is one forward pass.: Exchangeability turns a TFM into a Gibbs kernel. For every i and every assignment y with $p(y_{-i} \mid X) > 0$,

$$p(y_i \mid y_{-i}, X) = p(y_i \mid x_i, D_{-i}), \qquad D_{-i} = \{(x_j, y_j)\}_{j \neq i} \tag{2}$$

(Lemma 1): the conditional of one latent label given all the others is the PPD of a query point x_i with the rest of the current assignment as its context. Multiplying by the weak-label factor gives the full conditional of π in (1), $\pi(y_i = k \mid y_{-i}) \propto \tilde{p}(k \mid x_i, D_{-i})\, \ell_i(k)$. A Gibbs sweep over latent labels therefore needs only forward passes of a frozen TFM, and the weak labels enter only through ℓ. A real TFM only approximates the PPD, so for it this is an approximate Gibbs kernel.

Blocks.: One forward pass per label would cost n passes per sweep. We instead partition all n points at random into K blocks in every sweep and, for each block B, query the TFM once with the context D_{-B} (the other blocks with their current labels). For PLL, each *free* y_i, $i \in B$ (a label not fixed by the weak labels, i.e. with at least two candidates), is drawn from $\propto \tilde{p}(\cdot \mid x_i, D_{-B})\, \ell_i(\cdot)$; blocks include the points whose label is fixed, so that the context keeps its composition. For LLP, blocks are groups of whole bags, and each bag's labels are drawn from $\prod_{i \in B_g} \tilde{p}(y_i \mid x_i, D_{-B})$ conditioned on its counts, exactly, by dynamic programming. An optional Metropolis–Hastings (MH) step accepts or rejects the whole block, which makes the update exact (Proposition S2) at one forward pass per changed label. Contexts above 2,000 points are subsampled uniformly.

Warm start and prediction.: A chain starts from a cheap estimate P_0 of the labels: for PLL, the TFM fitted on one candidate per point chosen uniformly at random, cross-fitted over 5 folds; for LLP, linear DLLP [3]. Initial labels are drawn from P_0 restricted by ℓ. After T sweeps, of which the first T_0 are discarded, the prediction for a test point is the Rao–Blackwellised average over S evenly spaced samples,

$$\hat{p}(y_* \mid x_*) = \frac{1}{S} \sum_{s=1}^{S} \tilde{p}(y_* \mid x_*, (X, y^{(s)})). \tag{3}$$

This *inductive* predictor is our default; in the *transductive* variant, an ablation, test points join the chain.

Several starts and several TFMs.: In the model of Section 5.2, which covers instance-level weak labels, where a chain ends can depend on where it starts; for LLP we have only empirical evidence of this (Appendices E and G). The *multi-start* sampler runs three chains that differ only in their initial labels and averages their predictions (3) with equal weights: the *warm* start above; a *learner* start, in which P_0 is the prediction of a specialised weak-label learner L tuned on the weak labels alone (DLLP with an MLP for LLP, CC [2] for PLL); and a *cold* start drawn from the normalised likelihood (for PLL, uniformly from each candidate set; for LLP, uniformly among the labellings that match each bag's counts). The *multi-TFM* sampler averages the multi-start predictions of several TFMs, again with equal weights. Neither average needs labels or a selection step. The TFMs stay frozen throughout, but the starts are not free of training. What is fitted, by configuration: the single chain fits nothing for PLL (its warm start uses the TFM itself) and a linear DLLP model for LLP; the multi-start and multi-TFM samplers add the learner L, tuned over 20 configurations on the weak labels; no configuration fits anything to true labels.

Cost and settings.: A chain costs $TK + S$ forward passes. All settings were fixed on development datasets before any evaluation and are shared by all datasets and studies: $K = 10$, $T = 30$, $T_0 = 10$, $S = 10$ (310 passes).

5 Theory

Proofs are in Appendix A.

5.1 Exact and approximate TFMs

Lemma 1 (Full conditionals). *Under the exchangeable prior, (2) holds for every i and every y with $p(y_{-i} \mid X) > 0$.*

Theorem 1 (Exactness). *Assume (A0)–(A3) and $\tilde{p} = p$. The single-site Gibbs chain with updates $\pi(y_i = k \mid y_{-i}) \propto \tilde{p}(k \mid x_i, D_{-i})\ell_i(k)$ for instance-level weak labels and, for LLP, the MH chain that swaps two differently labelled points of a bag are reversible with respect to π and irreducible on its support. The inductive estimator (3) then converges almost surely to $\sum_y \pi(y)\, p(y_* \mid x_*, (X, y))$, which equals $p(y_* \mid x_*, X, W)$ when the feature distribution carries no information about $p(y \mid x)$ under the prior (in general up to a TV term, Appendix A); the transductive chain, in which x_* joins as a free point, converges to $p(y_* \mid x_*, X, W)$ exactly.*

With an exact TFM the exact variants of In-Context Gibbs thus compute the Bayes-optimal predictor under the TFM's prior (Proposition S1). A real TFM is only approximately a PPD, and its conditionals need not be compatible with any joint distribution. Samplers built from possibly incompatible conditionals are classical [34, 41], and their stationary distribution depends on the scan order [42]. The next result bounds how far it is from the posterior.

Theorem 2 (Stability under an approximate TFM). *Let P be the kernel of an exact variant with the exact PPD (stationary distribution π) and $\tilde{P}$ the same kernel with a TFM $\tilde{p} > 0$. Then $\tilde{P}$ has a unique stationary distribution $\tilde{\pi}$, and $\mathrm{TV}(\tilde{\pi}, \pi) \le \tau(P)\, \bar{\varepsilon}$, where $\tau(P) = \sum_{t \ge 0} \vartheta(P^t)$ with ϑ Dobrushin's ergodicity coefficient and $\bar{\varepsilon} = \sum_y \tilde{\pi}(y)\, \mathrm{TV}(\tilde{P}(y, \cdot), P(y, \cdot))$.*

The bound is (integrated mixing time of the exact chain) $\times$ (error per step), a classical form [43–46], here non-asymptotic and with MH steps included.

Uncorrected blocks.: A block update is exact when its proposal, the product of the TFM's predictives over the block, is accepted with an MH ratio (Proposition S2). The default sampler skips that step, and for it the bound above is useless: on exactly solvable problems it exceeds 1 in most cases, while the true $\mathrm{TV}(\tilde{\pi}, \pi)$ is small but does not shrink with n (Section A.5). Predictions behave better. In a conjugate model with unconstrained unknown labels and a fresh random block at every step, the stationary mean of the unknown-label counts equals their exact posterior mean under a prior concentration inflated by $K(n-1)/((K-1)n)$, and the resulting bias of the predictor is of order $1/((K-1)N)$ for N labelled points (Proposition S3). For the partition scan that the sampler uses this holds to leading order numerically, and for other weak labels, bag-blocks in particular, we have numerical evidence only.

5.2 The sampler as run: response maps and basins

The next result does not assume that the TFM is close to a PPD. It concerns the label accuracy of a chain of Algorithm 1 as run on instance-level weak labels: uncorrected blocks, a fresh random partition in every sweep, any deterministic TFM, fixed weak-label factors, context not subsampled. It does not cover LLP, whose bag updates are not independent draws. Let F be the set of points whose label is unknown, $n_F = |F|$, $\rho = n_F/n$, and y° the true labels, which only the analysis uses. The *label accuracy* is $r(y) = n_F^{-1} \sum_{i \in F} \mathbb{1}[y_i = y_i^\circ]$, and R_s is its value after sweep s. For a block B, $c_i(y, B)$ is the probability that the redraw of $i \in B \cap F$ is correct, and $r_{-B}(y)$ the accuracy on $F \setminus B$, the part of the context the TFM sees.

Assumption 1 (Homogeneous response, (H_δ))**.** There are a function $\Phi : [0,1] \to [0,1]$ (the *response map*), fixed offsets h_i with $|h_i| \le h_{\max}$ and $\sum_{i\in F} h_i = 0$, and $\delta \ge 0$ such that for every state y and every block B of $\lfloor n/K \rfloor$ or $\lceil n/K \rceil$ points with $B \cap F \neq \emptyset$ and $F \not\subseteq B$,

$$\Big| \frac{1}{|B \cap F|} \sum_{i \in B\cap F} \big(c_i(y, B) - h_i\big) - \Phi\big(r_{-B}(y)\big) \Big| \le \delta.$$

(H_δ^-) keeps only the lower bound ($\ge \Phi - \delta$), (H_δ^+) only the upper bound.

The response map gives the accuracy of the TFM's relabelling of a block as a function of the accuracy of its context: it depends on *how many* context labels are right, up to per-point difficulty h_i and a tolerance δ for *which* ones are right. This is a strong, uniform requirement, which we treat as a model of the TFM to be confronted with data; Φ need not be monotone or continuous. Let $\underline{\Phi}(x) = \inf\{\Phi(r) : r \in [0,1], r \ge x\}$, $\overline{\Phi}(x) = \sup\{\Phi(r) : r \in [0,1], r \le x\}$, and for margins $\nu, \lambda > 0$ let $p(\nu,\lambda) = e^{-2n_F\lambda^2} + 3K\, e^{-c\rho n_F \nu^2}$ with $c = a^2 \min\{\frac{2}{9}, \frac{1}{18h_{\max}^2}\}$ and $a = 1 - \lceil n/K \rceil / n_F$.

Theorem 3 (Floors, ceilings and basins)**.** *Run Algorithm 1 in the setting above with $n_F > \lceil n/K \rceil$, write $p = p(\nu,\lambda)$, and let $H \ge 1$.*

- (a) Floor. *Under (H_δ^-), if $\underline{\Phi}(x-\nu) - \delta - \lambda \ge x$ and $R_s \ge x$, then $R_{s'} \ge x$ for all $s < s' \le s + H$ with probability at least $1 - Hp$, and the first sweep τ at which the chain is below x satisfies $\mathbb{E}[\tau - s] \ge 1/p$.*
- (b) Ceiling. *Under (H_δ^+), if $\overline{\Phi}(x+\nu) + \delta + \lambda \le x$ and $R_s \le x$, then $R_{s'} \le x$ for all $s < s' \le s + H$ with probability at least $1 - Hp$.*
- (c) Progress. *Under (H_δ^-), if, for some levels $r_0 < b \le 1$ and some $\gamma > 0$, $\underline{\Phi}(x - \nu) - \delta - \lambda \ge x + \gamma$ for all $x \in [r_0, b)$, then a chain with $R_0 \ge r_0$ satisfies $R_J \ge b$ for some $J \le \lceil (b - r_0)/\gamma \rceil$ with probability at least $1 - Jp$.*

For a non-decreasing Φ, read this with the *gain* $g(r) = \Phi(r) - r$. Where g exceeds a margin (set by δ, λ and the variation of Φ over ν), the chain climbs (c) and does not fall back (a); where g is below minus that margin, it does not climb (b). If g is negative on an interval and positive above it, the model has two basins: a chain that starts sufficiently far above the crossing stays up, one that starts sufficiently far below stays down, and the expected time to leave is at least $1/p$ sweeps, which for a fixed fraction of unknown labels and fixed margins grows exponentially with n_F. In this model more sweeps do not repair a start in the wrong basin, which motivates the multi-start sampler. Real chains also mix slowly: run for 100 instead of 30 sweeps they kept improving (+0.053 on average, most with TabPFN-3; Section 6.5), which the theorem does not model. The theorem describes a mechanism and its scaling; its constants are loose (Section A.6).

When can the sampler help?: The naive use of a TFM conditions it on one label per point drawn from the weak labels. When the naive label depends on the features only through the true label, through a channel with equal off-diagonal entries in each column (positive-unlabelled (PU) data with labels selected completely at random, binary label noise, semi-supervised learning, PLL with uniform independent distractors, LLP with random bags), the naive target ranks points within each class as the true posterior does (Proposition S5). For PU, label noise and semi-supervised learning the naive labels are the observation: nothing is lost by them, and the sampler can only add the bias of uncorrected blocks. For PLL they carried less Fisher information than the candidate sets in every case we computed, and for LLP a learner that treats them as independent point labels uses about 3% of the information in the counts of bags of 8 (both for a logistic model). Only for PLL and LLP, then, can the sampler recover information through ℓ, which predicted, before evaluation, where In-Context Gibbs would beat the naive TFM (Appendix M).

6 Experiments

6.1 Design: three pre-registered studies

Settings and baselines.: The main claims concern LLP with random bags of 8 (ROC AUC) and PLL in which each wrong class enters the candidate set independently with probability $q \in \{0.3, 0.6\}$ (accuracy). Weak labels are generated on the training part of stratified 70/30 splits (seeds 0–2) shared by all methods; datasets have 500 to 10,000 rows. Studies 1 and 2 run the peer-reviewed baselines of each setting, in their published defaults and tuned by 20 random configurations scored on a validation split *with weak labels only*: for LLP, DLLP [3] with an MLP or a linear model, LLP-VAT [4], EasyLLP [12] and GLWS [5]; for PLL, PRODEN [1], GLWS/ILL [5, 19], RC and CC [2]. With the same TFM we also run the *naive TFM* (one label per point consistent with the weak labels), *hard-EM* (arg max instead of sampling) and an *oracle* (the TFM given the true labels). In-Context Gibbs is never tuned per dataset. A rule written before the evaluation runs fixed the *primary comparators*: the peer-reviewed baseline with the highest mean score, after tuning, on development datasets set aside for choosing settings; it selected linear DLLP in its default configuration and tuned CC. In Study 3 a *cell* is one run (TFM, dataset, seed), a *pair* a (TFM, dataset) averaged over seeds, and differences below 0.001 count as ties in W/T/L counts (Studies 1 and 2 count exact ties only); the Wilcoxon tests use every non-zero difference.

Studies 1 and 2: one chain against specialised learners.: H2 (LLP) and H4 (PLL) require the single chain to beat the primary comparator and the naive TFM (two-sided Wilcoxon signed-rank over datasets, Holm-adjusted) *and* to beat the baseline with the best mean score on ≥70% of datasets (LLP), or to match or beat it on ≥60% at each level (PLL); H3 compares it with hard-EM on log loss [47]. *Study 1* used 24 OpenML-CC18 datasets [48] with TabPFN v2 and, as a registered ablation, TabICL v2. *Study 2* replicated it on 28 new datasets from curated OpenML suites (OpenML100, the AutoML benchmark, the suites of 49, TabZilla, TabRepo and TabArena) with TabICL v2, changing nothing else (H2′–H4′).

Study 3: several starts and several TFMs.: *Study 3* used 20 further datasets (12 binary with LLP, 8 multiclass with PLL at $q = 0.6$) from the OpenML suites of Study 2, among them TabArena [50], and OpenML-CC18, excluding every dataset used earlier and TabDPT's pretraining sets. It ran TabICL v2, TabPFN-3 [21] and TabDPT [22] on all datasets and TabPFN v2 on the 11 within its published limits

(at most 500 features and 10 classes). Per dataset, Δ_1 is the mean over TFMs of (multi-start − single chain), and Δ_2 is the multi-TFM score minus the mean over TFMs of the multi-start score, the expected score of a TFM picked at random. H1″: $\Delta_1 > 0$; H2″: $\Delta_2 > 0$ (one-sided Wilcoxon signed-rank, Holm-adjusted). Three criteria were registered without a test. H3″: a diagnostic computed without labels identifies the cells in which the single chain fails, i.e. is more than 0.02 below the better of the multi-start average and L (AUC ≥ 0.80). C4″: the multi-start sampler is more than 0.02 below the better of the single chain and the tuned learner L in at most 10% of (TFM, dataset) pairs. C5″: the multi-TFM average is on average within 0.01 of the best TFM of each dataset and more than 0.02 behind it on at most 3. The only baseline registered in Study 3 is L (tuned DLLP-MLP and tuned CC, the strongest LLP baseline and the primary PLL comparator of Study 2, both neural or linear); Section 6.5 adds non-neural baselines. All protocols were committed before the evaluation data were downloaded; development between studies was exploratory (Appendices B and G).

6.2 Study 2: one chain with TabICL v2 outperformed specialised neural and linear learners

In Study 2 all pre-registered tests passed (Table 1, right). On LLP the single chain improved AUC over the primary comparator by +0.052 [+0.024, +0.086] (16/0/2) and over the strongest LLP baseline, tuned DLLP (MLP), by +0.041 [+0.018, +0.072] (better on 16/18 datasets). On PLL it improved accuracy over tuned CC by +0.012 at $q = 0.3$ and +0.012 at $q = 0.6$, a small gain close to the oracle ceiling (0.8 and 1.9 points below it), and was at least as accurate as the strongest PLL baseline on 10/10 and 9/10 datasets (a rerun with a second chain is in Table S26). With the same TFM and warm start, sampling beats taking the arg max: In-Context Gibbs had lower log loss than hard-EM in 78.9% of 38 (dataset, setting) comparisons (registered H3′).

6.3 Study 1 and the role of the TFM

Study 1 used TabPFN v2 (Table 1, left). In-Context Gibbs improved on the naive TFM by +0.148 AUC on LLP and +0.099 and +0.180 accuracy on PLL, and at $q = 0.6$ it beat tuned CC on 7 of the 10 datasets with fewer than eight classes (+0.033 on average). On the four datasets with at least eight classes, TabPFN v2's chains stalled (Table S4), while with TabICL v2 every chain ended at a label accuracy of 0.925 or above and the registered ablation passed every test (Appendix C); Study 2 then tested TabICL v2 prospectively.

Figure 1 shows why the TFM matters: the gain $\Phi(r) - r$, measured by redrawing every block given an assignment with a fraction r of correct labels and uniformly placed errors, is positive throughout with TabICL v2 and negative for intermediate r with TabPFN v2 (Appendix G relates these maps to the chains).

6.4 Several starts and several TFMs

Confirmed.: Both hypotheses of Study 3 passed (Table 2; means by TFM in Table S10, per dataset in Figure S3). Differences pool AUC on LLP datasets and accuracy on PLL datasets. Averaging three starts improved on the single chain by +0.058 on average (13/6/1 datasets), most on PLL (+0.136; LLP +0.007) and where single chains stall (TabPFN-3 on PLL: 0.371 to 0.650). The multi-TFM average improved on a TFM picked at random by +0.046 (18/2/0) and was level with the best TFM of each dataset (mean gap 0.000). Both tests also pass without the 5 datasets flagged before the study for group or sequence structure (registered sensitivity analysis; $p_{\text{Holm}} = 0.003$ and <0.001). Split by setting, both differences are positive in both settings (Table S15). Appendix E reports the remaining registered criteria.

Exploratory analysis.: The following was not registered and was computed after the confirmatory scoring (Appendix E). (i) The multi-TFM average was above L on all 20 datasets, by +0.064 on average [+0.030, +0.110], and it was never more than 0.02 below the better of a TFM's single chain and L (0 of 71 pairs, the measure of C4″); the average of single chains over TFMs, which does not use L, was above L on all 12 LLP datasets. (ii) The benefit of several starts comes from the learner start: the chain started from L alone matches the three-start average (+0.001), the cold start adds little, and averaging the single chain with L's own prediction, with no further chain, does as well (+0.008, $p = 0.26$). (iii) The three chains ended within 0.03 of each other in label accuracy in 81% of cells (91% on LLP, 61% on PLL); the PLL gain of the multi-start average is +0.328 in the other cells and +0.006 where chains agree. (iv) The advantage over L is not only that of averaging with L: averaging the single chains of all TFMs with L's prediction, which uses the same information without chains started from L, was above L (+0.037, 20/0/0), and the multi-TFM average was above that average in turn (+0.027, 16/0/4).

6.5 Supplementary analysis registered after Study 3

After Study 3 had been scored we registered three further analyses, with a scoring script, and ran them on the stored splits and weak labels of Study 3 (Appendix F).

Non-neural baselines.: We added self-training with gradient-boosted trees (pseudo-labels restricted to the candidate sets for PLL, to each bag's count for LLP; one fixed configuration) and, for PLL, the k-nearest-neighbour vote over candidate sets of Hüllermeier and Beringer [14] (PL-kNN, $k = 10$). On label proportions the multi-TFM average was ahead of tree-based self-training on all 12 datasets (median +0.017 AUC), and on partial labels it was ahead of PL-kNN on all 8 datasets by +0.188 (p_{Holm}=0.016, two-sided).

Naive TFM and hard-EM with four TFMs.: Averaged over the TFMs of each dataset (Table S17), the single chain improved on the naive TFM by +0.128 (18/1/1) and had lower log loss than hard-EM on 17/0/3 datasets (two-sided $p<0.001$), as in Studies 1 and 2.

Longer chains.: Rerun for 100 instead of 30 sweeps on all 72 PLL cells of TabICL v2, TabPFN-3 and TabDPT, chains kept improving: test accuracy rose by +0.053 on average, most with TabPFN-3 (+0.111), whose label accuracy rose from 0.40 to 0.49 (Table S18).

6.6 What to run

Table 3 collects, on the datasets of Study 3, the score, the cost and the strength of evidence of each configuration, ordered

TABLE 1
Studies 1 and 2, pre-registered tests of the single chain. Δ: mean over datasets of (In-Context Gibbs − comparator), 3 seeds each; W/T/L: datasets won/tied/lost; p_{Holm}: two-sided Wilcoxon signed-rank, Holm-adjusted within each hypothesis. Bold: significant in favour of In-Context Gibbs. The two PLL rows against CC coincide only after rounding (Appendix D).

			Study 1: TabPFN v2			Study 2: TabICL v2, new datasets		
Setting	Metric	Gibbs vs.	Δ	W/T/L	p_{Holm}	Δ [95% CI]	W/T/L	p_{Holm}
LLP, bag size 8	AUC	DLLP (linear)	+0.024	8/0/2	0.232	+0.052 [+0.024, +0.086]	16/0/2	**<0.001**
		Naive TFM	+0.148	10/0/0	**0.004**	+0.116 [+0.085, +0.150]	18/0/0	**<0.001**
PLL, $q = 0.3$	Acc.	CC (tuned)	+0.022	10/0/4	0.084	+0.012 [+0.006, +0.019]	8/2/0	**0.016**
		Naive TFM	+0.099	13/0/1	**<0.001**	+0.022 [+0.012, +0.033]	10/0/0	**0.008**
PLL, $q = 0.6$	Acc.	CC (tuned)	−0.132	7/0/7	0.715	+0.012 [+0.006, +0.019]	8/1/1	**0.016**
		Naive TFM	+0.180	12/0/2	**0.002**	+0.205 [+0.094, +0.337]	10/0/0	**0.008**
Pre-registered verdict			H2: not met; H4: not met			H2′: met; H4′: met		

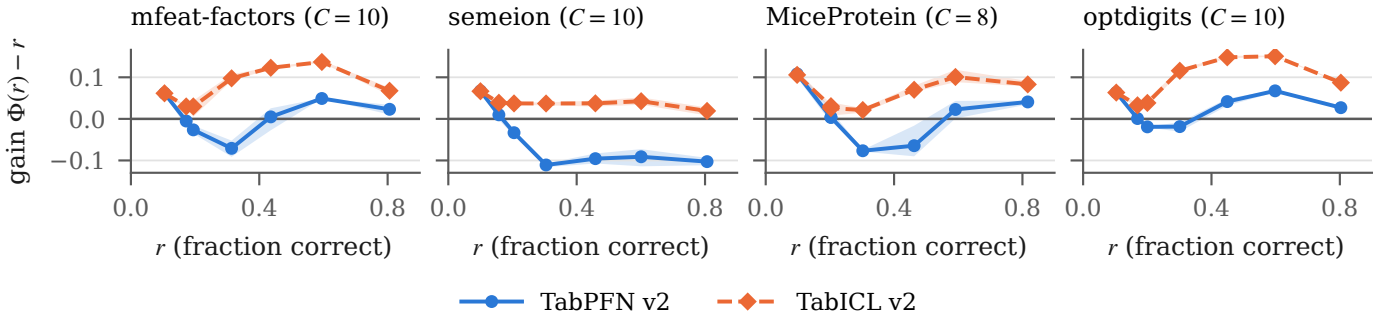


Fig. 1. Response maps for PLL at $q = 0.6$ on the four Study 1 datasets with at least eight classes: gain $\Phi(r) - r$ when every block is redrawn given an assignment with a fraction r of correct labels and uniformly placed errors (needs true labels); mean over 3 seeds, with the range over seeds shaded. On optdigits the context is subsampled, outside the scope of Theorem 3.

TABLE 2
Study 3 (20 new datasets, four TFMs, 3 seeds): registered hypotheses and criteria (Section 6.1). Differences pool AUC (LLP datasets) and accuracy (PLL datasets): mean over datasets [bootstrap 95% interval], W/T/L, one-sided Wilcoxon signed-rank p, Holm-adjusted.

	Statement	Result	Outcome
H1″	multi-start > single chain	+0.058 [+0.016, +0.114], 13/6/1, p<0.001	**confirmed**
H2″	multi-TFM > a TFM drawn at random	+0.046 [+0.019, +0.079], 18/2/0, p<0.001	**confirmed**
H3″	diagnostic AUC ≥ 0.80	0.766 [0.59, 0.92]	not confirmed
C4″	regressions $\leq 10\%$	10/71 (14.1%)	not met
C5″	multi-TFM $\approx$ best TFM	gap 0.000; 1/20 worse by > 0.02	met

by cost; rows marked exploratory were computed after scoring from the stored predictions. For label proportions, averaging single chains over TFMs, without any tuned learner, reached 0.823 AUC. For partial labels the gain comes from starting chains from a trained learner: starting one chain per TFM from it and averaging matched the multi-TFM average (−0.001, 5/7/8) at a third of its cost, and within one TFM a single chain averaged with the learner's prediction did as well as three starts (+0.008); both are exploratory. The registered multi-TFM average, which has the strongest evidence, costs 9 or 12 chains and a tuned learner.

6.7 Real weak labels and cost

On three real partial-label datasets (BirdSong, Lost, MSRCv2; 10 random 90/10 splits; single chain with TabICL v2, which Study 1 registered for these data because they exceed TabPFN v2's limit of 10 classes), In-Context Gibbs had the highest mean accuracy on all three and the best mean rank among 10 methods (2.15; Table S29). On one T4 GPU a chain took a median of 41 s in Study 2, and the 9 or 12 chains of the multi-TFM average a median of 37 minutes per dataset and seed in Study 3, not counting the tuning of L.

7 Limitations

The comparison with specialised baselines is confirmatory only for the single chain with TabICL v2 (Study 2); the non-neural baselines of Section 6.5 were run untuned. Theorem 3 covers the default sampler only for instance-level weak labels with contexts that are not subsampled, and its constants are loose. Weak labels are synthetic except for three PLL datasets,

TABLE 3
Score, cost and evidence of each configuration on the 20 datasets of Study 3 (means over datasets; single-TFM rows averaged over the TFMs of each dataset), ordered by cost. A chain costs 310 forward passes; the warm start adds 5 for PLL, where it cross-fits the TFM. A chain took a median of 92 s to 208 s on one T4 GPU in Study 3. All configurations except L, GBDT and PL-kNN also need the TFM itself; "tuned L" adds 20 tuning runs. "Exploratory": computed after scoring, without error control.

Configuration	LLP, AUC	PLL, acc.	TFM passes (LLP / PLL)	Trained parts	Evidence
Naive TFM	0.718	0.393	1 / 1	none	registered supplement
Hard-EM with the TFM	0.800	0.539	101 / 106	LLP start: linear DLLP	registered supplement
Single chain	0.801	0.589	310 / 315	LLP start: linear DLLP	confirmatory with TabICL v2 only (Study 2)
Single chain from L	0.799	0.740	1 chain	tuned L	exploratory
Single chains, averaged over TFMs	0.823	0.747	3–4 chains	LLP start: linear DLLP	exploratory
Single chains over TFMs, averaged with L	0.820	0.752	3–4 chains	tuned L; LLP: also linear DLLP	exploratory
Chains from L, averaged over TFMs	0.823	0.814	3–4 chains	tuned L	exploratory
Multi-start	0.807	0.725	3 chains	tuned L; LLP: also linear DLLP	confirmatory vs. single chain
Multi-start, TabICL v2 only	0.803	0.805	3 chains	tuned L; LLP: also linear DLLP	exploratory (TFM chosen after Study 1)
Multi-TFM average	0.828	0.808	9–12 chains	tuned L; LLP: also linear DLLP	confirmatory vs. a TFM drawn at random
Tuned learner L	0.783	0.717	none (20 tuning runs)	tuned on weak labels	Study 3 reference
GBDT self-training	0.796	0.519	none (CPU)	fixed configuration	registered supplement
PL-kNN	–	0.620	none (CPU)	none	registered supplement

and the claims cover LLP with random bags of 8 and PLL with uniform distractors at $q \leq 0.6$. Detecting a stalled chain without labels remains open.

8 Conclusion

Under exchangeability, the predictive of an exact TFM is the full conditional of a Gibbs sampler over unknown labels, and a real TFM approximates it. In-Context Gibbs turns this into one procedure for candidate sets and label proportions. In Study 2 a single chain with TabICL v2 outperformed the specialised neural and linear learners we ran on new datasets. Averaging over starts improved on single chains, and averaging over TFMs improved on a TFM drawn at random; in a registered supplement the multi-TFM average also beat untuned non-neural baselines. Whether a chain stalls depends on the TFM and on where the chain starts. Open problems: a label-free diagnostic of stalling, a cheaper way to escape a wrong start than averaging over many chains (starting chains from a trained learner was enough in our exploratory analysis), response-map bounds with usable constants, and feature-dependent weak labels.

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References

[1] J. Lv, M. Xu, L. Feng, G. Niu, X. Geng, and M. Sugiyama, "Progressive identification of true labels for partial-label learning," in *Proceedings of the 37th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 119, 2020, pp. 6500–6510.

[2] L. Feng, J. Lv, B. Han, M. Xu, G. Niu, X. Geng, B. An, and M. Sugiyama, "Provably consistent partial-label learning," in *Advances in Neural Information Processing Systems*, vol. 33, 2020.

[3] E. M. Ardehaly and A. Culotta, "Co-training for demographic classification using deep learning from label proportions," in *IEEE International Conference on Data Mining Workshops (ICDMW)*, 2017, pp. 1017–1024.

[4] K.-H. Tsai and H.-T. Lin, "Learning from label proportions with consistency regularization," in *Proceedings of the 12th Asian Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 129, 2020, pp. 513–528.

[5] H. Chen, J. Wang, L. Feng, X. Li, Y. Wang, X. Xie, M. Sugiyama, R. Singh, and B. Raj, "A general framework for learning from weak supervision," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235, 2024, pp. 7462–7485.

[6] N. Hollmann, S. Müller, K. Eggensperger, and F. Hutter, "TabPFN: A transformer that solves small tabular classification problems in a second," in *International Conference on Learning Representations*, 2023.

[7] N. Hollmann, S. Müller, L. Purucker, A. Krishnakumar, M. Körfer, S. B. Hoo, R. T. Schirrmeister, and F. Hutter,

"Accurate predictions on small data with a tabular foundation model," *Nature*, vol. 637, no. 8045, pp. 319–326, 2025.

[8] J. Qu, D. Holzmüller, G. Varoquaux, and M. Le Morvan, "TabICL: A tabular foundation model for in-context learning on large data," in *Proceedings of the 42nd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 267, 2025, pp. 50 817–50 847.

[9] ——, "TabICLv2: A better, faster, scalable, and open tabular foundation model," in *International Conference on Machine Learning*, 2026, arXiv:2602.11139.

[10] S. Müller, N. Hollmann, S. Pineda Arango, J. Grabocka, and F. Hutter, "Transformers can do Bayesian inference," in *International Conference on Learning Representations*, 2022.

[11] H. Wang, R. Xiao, Y. Li, L. Feng, G. Niu, G. Chen, and J. Zhao, "PiCO: Contrastive label disambiguation for partial label learning," in *International Conference on Learning Representations*, 2022.

[12] R. Busa-Fekete, H. Choi, T. Dick, C. Gentile, and A. M. Medina, "Easy learning from label proportions," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[13] S. Havaldar, N. Sharma, S. Sareen, K. Shanmugam, and A. Raghuveer, "Learning from label proportions: Bootstrapping supervised learners via belief propagation," in *International Conference on Learning Representations*, 2024.

[14] E. Hüllermeier and J. Beringer, "Learning from ambiguously labeled examples," *Intelligent Data Analysis*, vol. 10, no. 5, pp. 419–439, 2006.

[15] M.-L. Zhang and F. Yu, "Solving the partial label learning problem: An instance-based approach," in *Proceedings of the Twenty-Fourth International Joint Conference on Artificial Intelligence*, 2015, pp. 4048–4054.

[16] S. Rüping, "SVM classifier estimation from group probabilities," in *Proceedings of the 27th International Conference on Machine Learning*, 2010, pp. 911–918.

[17] F. Yu, D. Liu, S. Kumar, T. Jebara, and S.-F. Chang, "$\propto$SVM for learning with label proportions," in *Proceedings of the 30th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 28, 2013, pp. 504–512.

[18] R. Busa-Fekete, T. Dick, C. Gentile, H. Kaplan, T. Koren, and U. Stemmer, "Nearly optimal sample complexity for learning with label proportions," *arXiv preprint arXiv:2505.05355*, 2025.

[19] H. Chen, A. Shah, J. Wang, R. Tao, Y. Wang, X. Li, X. Xie, M. Sugiyama, R. Singh, and B. Raj, "Imprecise label learning: A unified framework for learning with various imprecise label configurations," in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 59 621–59 654.

[20] V. Shukla, Z. Zeng, K. Ahmed, and G. Van den Broeck, "A unified approach to count-based weakly supervised learning," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[21] L. Grinsztajn, K. Flöge, O. Key, F. Birkel, P. Jund, B. Roof, M. Manium, S. B. Hoo, M. Bühler, A. Garg, D. Safaric, J. Robertson, B. Jäger, S. Alessi, A. Hayler, V. Moroshan, L. Purucker, P. Singer, A. Arazi, J. Siems, J. H. Metzen, G. Grab, N. Erickson, S. Guo, E. Kalfon, S. Bing, D. Salinas, C. Cornu, L. C. Wehrhahn, D. Kriuchkova, K. Kaya, L. Sidhoum, M. Salmon, J. Chen, M. Hulsebos, Y. LeCun, S. Müller, B. Schölkopf, S. Gambhir, N. Hollmann, and F. Hutter, "TabPFN-3: Technical report," arXiv:2605.13986, 2026.

[22] J. Ma, V. Thomas, R. Hosseinzadeh, A. Labach, J. C. Cresswell, K. Golestan, G. Yu, A. L. Caterini, and M. Volkovs, "TabDPT: Scaling tabular foundation models on real data," in *Advances in Neural Information Processing Systems*, vol. 38, 2025.

[23] G. Kang, C. J. Lee, and X. Cheng, "Transformers can learn posterior predictive distributions in-context," in *Proceedings of the 43rd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 306, 2026, pp. 55 565–55 596.

[24] J. Hu and M. Ghelichi, "Noise immunity in in-context tabular learning: An empirical robustness analysis of TabPFN's attention mechanisms," *arXiv preprint arXiv:2604.04868*, 2026.

[25] Y. Li, X. Chang, M. Kara, X. Liu, A. Roy-Chowdhury, and S. Oymak, "When and how unlabeled data provably improve in-context learning," in *Advances in Neural Information Processing Systems*, vol. 38, 2025.

[26] S. Liu, Y. Chang, M. Cheng, Q. Tian, and P. Li, "In-context positive-unlabeled learning," *arXiv preprint arXiv:2605.05591*, 2026.

[27] J. Fan, P. Rosu, A. T. Wang, Z. M. Li, L. Carin, and X. Cheng, "In-context semi-supervised learning," *arXiv preprint arXiv:2512.15934*, 2025.

[28] A. Möllers, M. Sextro, J. Hense, G. Dernbach, and K.-R. Müller, "In-context multiple instance learning," *arXiv preprint arXiv:2606.06458*, 2026.

[29] J. Sepin, "Multiple imputation of a continuous outcome with fully observed predictors using TabPFN," *Stats*, vol. 9, no. 2, p. 38, 2026.

[30] J. Feitelberg, D. Saha, K. Choi, Z. Ahmad, A. Agarwal, and R. Dwivedi, "TabImpute: Universal zero-shot imputation for tabular data," *arXiv preprint arXiv:2510.02625v4*, 2026.

[31] T. Nagler and D. Rügamer, "Uncertainty quantification for prior-data fitted networks using martingale posteriors," *arXiv preprint arXiv:2505.11325*, 2025.

[32] K. Ng, E. Fong, D. T. Frazier, J. Knoblauch, and S. Wei, "TabMGP: Martingale posterior with TabPFN," *arXiv preprint arXiv:2510.25154*, 2025.

[33] J.-Q. Zhu and T. L. Griffiths, "Eliciting the priors of large language models using iterated in-context learning," *arXiv preprint arXiv:2406.01860*, 2024.

[34] J. Liu, A. Gelman, J. Hill, Y.-S. Su, and J. Kropko, "On the stationary distribution of iterative imputations," *Biometrika*, vol. 101, no. 1, pp. 155–173, 2014.

[35] H. Wu, H. Zimmermann, E. Sennesh, T. A. Le, and J.-W. van de Meent, "Amortized population Gibbs samplers with neural sufficient statistics," in *Proceedings of the 37th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 119, 2020, pp. 10 421–10 431.

[36] M. J. Johnson, J. Saunderson, and A. Willsky, "Analyzing Hogwild parallel Gaussian Gibbs sampling," in

Advances in Neural Information Processing Systems 26, 2013.

[37] C. De Sa, C. Re, and K. Olukotun, "Ensuring rapid mixing and low bias for asynchronous Gibbs sampling," in *Proceedings of the 33rd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 48, 2016, pp. 1567–1576.

[38] G. Celeux and J. Diebolt, "The SEM algorithm: a probabilistic teacher algorithm derived from the EM algorithm for the mixture problem," *Computational Statistics Quarterly*, vol. 2, pp. 73–82, 1985.

[39] M. A. Tanner and W. H. Wong, "The calculation of posterior distributions by data augmentation," *Journal of the American Statistical Association*, vol. 82, no. 398, pp. 528–540, 1987.

[40] S. Balakrishnan, M. J. Wainwright, and B. Yu, "Statistical guarantees for the EM algorithm: From population to sample-based analysis," *The Annals of Statistics*, vol. 45, no. 1, pp. 77–120, 2017.

[41] D. Heckerman, D. M. Chickering, C. Meek, R. Rounthwaite, and C. Kadie, "Dependency networks for inference, collaborative filtering, and data visualization," *Journal of Machine Learning Research*, vol. 1, pp. 49–75, 2000.

[42] S.-H. Chen and E. H. Ip, "Behaviour of the Gibbs sampler when conditional distributions are potentially incompatible," *Journal of Statistical Computation and Simulation*, vol. 85, no. 16, pp. 3266–3275, 2015.

[43] P. J. Schweitzer, "Perturbation theory and finite Markov chains," *Journal of Applied Probability*, vol. 5, no. 2, pp. 401–413, 1968.

[44] C. D. Meyer, Jr., "The condition of a finite Markov chain and perturbation bounds for the limiting probabilities," *SIAM Journal on Algebraic and Discrete Methods*, vol. 1, no. 3, pp. 273–283, 1980.

[45] E. Seneta, "Perturbation of the stationary distribution measured by ergodicity coefficients," *Advances in Applied Probability*, vol. 20, no. 1, pp. 228–230, 1988.

[46] G. E. Cho and C. D. Meyer, "Comparison of perturbation bounds for the stationary distribution of a Markov chain," *Linear Algebra and its Applications*, vol. 335, no. 1–3, pp. 137–150, 2001.

[47] A. Benavoli, G. Corani, J. Demšar, and M. Zaffalon, "Time for a change: a tutorial for comparing multiple classifiers through Bayesian analysis," *Journal of Machine Learning Research*, vol. 18, no. 77, pp. 1–36, 2017.

[48] B. Bischl, G. Casalicchio, M. Feurer, P. Gijsbers, F. Hutter, M. Lang, R. G. Mantovani, J. N. van Rijn, and J. Vanschoren, "OpenML benchmarking suites," in *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, vol. 1, 2021.

[49] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on typical tabular data?" in *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, vol. 35, 2022.

[50] N. Erickson, L. Purucker, A. Tschalzev, D. Holzmüller, P. Desai, D. Salinas, and F. Hutter, "TabArena: A living benchmark for machine learning on tabular data," in *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, vol. 38, 2025.